

Daily PM_{2.5} Estimation over Greater London: A Retrospective Study of Boosting, U-Net and Regional Pollution Correction

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Abstract

This study evaluates daily fine particulate matter (PM_{2.5}) estimation over Greater London using an existing multi-source data pipeline and a common retrospective benchmark. Regulatory and community-monitor observations are combined with meteorological, satellite, geographical, calendar and prior-day pollution predictors on a 1 km grid. The prepared 2021–2024 package contains 444,202 station-days, while final fitting uses 203,497 observed grid-cell days from 2021–2023. Histogram gradient boosting (HGB), constrained HGB, XGBoost, a multilayer perceptron, a standalone U-Net and an HGB–residual-U-Net hybrid are evaluated on a common benchmark of 9,619 London Air Quality Network station-days from 1 January to 14 December 2024. The original hybrid obtains RMSE 3.7765 $\mu\text{g m}^{-3}$, compared with HGB’s 3.7920 $\mu\text{g m}^{-3}$ and the standalone U-Net ensemble’s 4.1227 $\mu\text{g m}^{-3}$. In a separate matched refit, five past-only history summaries reduce RMSE from 3.7815 to 3.7679 $\mu\text{g m}^{-3}$. A subsequent day-level Ridge correction uses same-day regional observations from the Automatic Urban and Rural Network (AURN), without retraining the neural models. Local reproduction from saved temporal predictions gives MAE 2.1343 $\mu\text{g m}^{-3}$, RMSE 3.2309 $\mu\text{g m}^{-3}$ and $R^2 = 0.6471$: a 14.25% RMSE reduction relative to the temporal hybrid. A descriptive seven-day-block interval for corrected-minus-control RMSE is $[-0.8294, -0.2373]$ $\mu\text{g m}^{-3}$. MAE on observations at least 25 $\mu\text{g m}^{-3}$ falls from 14.0490 to 10.2219 $\mu\text{g m}^{-3}$, although 213 of 216 such readings remain underpredicted. The improvement reflects additional same-day regional information, not demonstrated architectural superiority or next-day forecast skill. Previously examined benchmark data, sparse spatial supervision and a documented wind-scaling defect constrain interpretation and model promotion.

Keywords: PM_{2.5}; Greater London; retrospective estimation; histogram gradient boosting; U-Net; residual learning; regional monitoring.

1 Introduction

Monitoring networks provide observations at specific locations, while a spatial pollution map requires estimates between those locations. The London Air Quality Network (LAQN) explicitly identifies this gap between monitoring coverage and spatial assessment [1]. A data-driven estimation system can combine measured pollution with meteorological and geographical information, but a dense output grid does not itself establish accuracy where no monitor is present. Evaluation must distinguish measured station values from estimated map values.

The present implementation makes it possible to examine whether spatial neural processing adds useful information beyond a strong tabular model. A U-Net can process neighbouring

predictor cells jointly, whereas a tabular boosting model operates on the feature vector of each queried cell. However, the tabular features already include spatially interpolated prior-day pollution. The incremental value of an additional spatial network is therefore an empirical question, rather than a consequence of architectural complexity.

This paper documents a Greater London pipeline and compares its completed model configurations under a common observation-level evaluation. Its contribution is an implementation-grounded case study of tabular, direct neural and hybrid residual estimation; no new model family or priority claim is made. The analysis addresses four questions: (i) how do these configurations compare on common regulatory observations; (ii) how large are the gains from a spatial correction and additional past-only temporal features; (iii) does additional same-day regional pollution information reduce episode errors; and (iv) what errors remain in spatial variation and high concentrations? Methods, source transformations and evaluation history are reported alongside the favourable and unfavourable results.

The task is retrospective daily estimation with rolling access to earlier pollution measurements. Models may use environmental covariates associated with target day t and measured London $\text{PM}_{2.5}$ from dates before t . The AURN-assisted extension additionally uses regional measurements on day t . Quarterly satellite composites can include imagery acquired later within the same quarter, and weather inputs are reanalysis products. Consequently, this study does not demonstrate a strict next-day operational forecast. Moreover, 2024 had been inspected in earlier experiments before the final comparison, so the benchmark is explicitly retrospective.

2 Related work and methodological context

Multi-source pollution estimation already has a substantial methodological foundation. For example, van Donkelaar et al. [2] combined satellite aerosol optical depth, chemical transport modelling and ground measurements to estimate monthly $\text{PM}_{2.5}$ and uncertainty globally. That study motivates the general use of complementary information sources, but its monthly global setting is different from this daily London benchmark. Its reported performance is therefore not used as a numerical baseline here. In particular, the present model’s Sentinel-5P aerosol index is not represented as an aerosol optical-depth measurement or a direct surface $\text{PM}_{2.5}$ retrieval.

Tree boosting provides a relevant baseline for heterogeneous environmental predictors. XGBoost is a scalable tree-boosting system [3]; the HGB implementation used here is the separate histogram-based regressor in scikit-learn [4]. The two are distinct implementations. This paper compares fitted configurations of these models and does not infer a universal ranking of their model families.

U-Net was introduced as an encoder–decoder architecture with skip connections for biomedical image segmentation [5]. The present implementation adapts that architectural pattern to continuous map regression, using Group Normalization [6] and squeeze-and-excitation blocks [7]. Label-density reweighting is conceptually related to work on imbalanced regression [8]; the implementation uses a specific capped, source-weighted histogram scheme. None of these architectural or weighting components is claimed as original to this study. The literature context is selective and does not establish that this implementation is the first London application or outperforms all published approaches.

3 Study area and data

3.1 Spatial and temporal representation

The model grid uses the British National Grid coordinate reference system (EPSG:27700) at 1 km spacing. Its tensor geometry is 48×64 , with 1,719 cells intersecting the adopted Greater London domain. The binary domain includes partial boundary cells; its cell count should not

Table 1: Data used by the final configurations. Source products and mathematical transformations are distinguished from directly measured $\text{PM}_{2.5}$.

Source	Representation and role
LAQN and Breathe London	Quality-controlled daily observations for sparse cell targets; previous-day interpolated fields for predictors.
ERA5	Hourly single-level reanalysis, reduced to daily and trailing three-day meteorological descriptors [11].
Sentinel-5P	Daily NO_2 , CO, aerosol index and cloud-fraction inputs, with availability, validation fraction and age channels [12].
Sentinel-2 L2A	Cloud-masked quarterly NDVI and NDBI composites and an availability channel [13].
ONS Census 2021	Population information processed into a spatial population-density predictor [14].
Ordnance Survey	Road coverage, greenspace and terrain descriptors from OpenMap Local, Open Greenspace and Terrain 50 [15].
LAEI 2022	Spatial emissions-inventory predictors retained as static inputs [16].
Grid and calendar	Domain masks, coordinates, day-of-year/day-of-week encodings and a time trend.
AURN (extension only)	Same-day regional measurements for a separate day-level correction; not new London training targets or U-Net channels [17].

be interpreted as the administrative area in square kilometres. The fractional-cell area in the prepared quality report is approximately $1,594.52 \text{ km}^2$. Predictor arrays cover 1 January 2021 through 31 December 2024, or 1,461 calendar days.

3.2 Observation sources and quality control

LAQN regulatory observations and Breathe London community observations provide the $\text{PM}_{2.5}$ supervision. LAQN is the primary evaluation source. The Breathe London Communities documentation describes a sensor network whose data are improved through correction to the reference network [9]; its API documentation identifies the data-access interface [10]. The two sources are consequently treated separately rather than assumed to have identical measurement properties or completely independent provenance.

The preparation code removes duplicate station-hours, retains hourly concentrations in the inclusive range $0\text{--}500 \mu\text{g m}^{-3}$, and requires at least 18 valid hourly values for a daily mean. Within each source, station-day values in the same grid cell are aggregated by their median. Let L_{tg} and B_{tg} denote the resulting LAQN and Breathe values for day t , cell g . The cell target is

$$y_{tg} = \begin{cases} (4L_{tg} + B_{tg})/5, & \text{both sources available,} \\ L_{tg}, & \text{LAQN only,} \\ B_{tg}, & \text{Breathe only.} \end{cases} \quad (1)$$

Cells without observations remain unlabelled. Equation 1 is an explicit aggregation of measurements, not a claim that mixed-source cells contain a purely regulatory target. Evaluation uses the individual LAQN station-day observations, rather than these mixed training targets.

The prepared package contains 444,202 station-days: 40,949 LAQN and 403,253 Breathe observations. Aggregation yields 294,364 observed grid-cell days over the complete package and 203,497 observed grid-cell days in the 2021–2023 final fitting window. A station-day and a grid-cell day are different units and are not used interchangeably.

3.3 Environmental predictors and missingness

The prepared cube has 71 channels, while the original final models retain 66. The five excluded channels are the stored fire-count, fire-radiative-power, local-fire and transported-smoke proxies. Their availability in the data directory does not mean they participate in the final models. The ordered retained feature list is provided in Appendix B.

ERA5 supplies meteorological covariates, including wind, temperature, dew point, pressure, precipitation, radiation, cloud cover and boundary-layer height. ERA5 is a reanalysis [11], so its values are not described as local sensor measurements. Satellite products are resampled to the common grid; the 1 km output spacing does not imply that all original satellite or weather inputs resolve independent 1 km variation.

The implementation records missing satellite retrievals through availability and related channels, permits limited forward filling of Sentinel-5P values for up to three days, and applies training-derived imputation where needed. Sentinel-2 uses quarterly median composites after cloud/scene masking. Continuous neural predictors are re-standardised for the fitting window from an existing clipped float16 feature cube. Re-standardisation cannot undo the original clipping or quantisation. Static masks and availability descriptors also make some coverage limitations explicit, including the retained OS tile coverage. Missing or imputed covariates are not presented as observed values.

3.4 Previous-day pollution predictors

The PM history fields use inverse-distance weighting (IDW) of observed cell targets from earlier dates. For a day with observations, up to eight nearest observed cells contribute, with weights proportional to $\max(d, 0.35)^{-2}$, where grid-cell distances correspond to kilometres on the 1 km grid. The resulting fields are shifted to form lag-one, lag-two and trailing-three-day-mean predictors. Nearest-observation distance, direct observation presence, availability and log observation count form four additional channels. Initial or completely missing-history cases use the preparation pipeline’s training-derived fallback median, with availability recorded separately.

These seven channels do not provide same-day measured $\text{PM}_{2.5}$ to the model. However, later 2024 estimates can depend on earlier 2024 observations. The word “prior” therefore describes the target-day information rule; it does not imply causal inference, removal of station-history effects, or independence between all train and evaluation locations.

3.5 Additional regional AURN observations

The regional extension uses DEFRA’s Automatic Urban and Rural Network, which provides hourly $\text{PM}_{2.5}$ measurements [17]. The implemented selection retains background or rural sites 30–110 km from central London. Of 18 selected sites, 15 supply usable observations in 2021–2024, yielding 16,413 valid station-days. Every one of the 1,461 dates has 6–15 reporting sites. The preparation retains ratified hourly values in $0\text{--}500\ \mu\text{g m}^{-3}$ and requires at least 18 valid hours per station-day. Reaggregating the shared hourly files reproduces the supplied daily values [18].

Daily regional features comprise an inverse-distance-weighted mean, maximum, a direction-weighted mean and $\log(1 + n_t)$ for reporting-site count n_t . The base station weight is $1/\max(d, 25)$ for distance d in kilometres. An availability field and three lag-one fields are also stored, but the selected same-day correction uses only the four features above. These inputs remain separate from the original London targets and 72-channel temporal U-Net input. The closest selected AURN station is 9.30 km from the 38 evaluated LAQN sites. This excludes obvious co-location in the selected set, without proving independence of every upstream measurement process.

The supplied direction-weighted feature has a known implementation defect: it computes the angle from standardised ERA5 wind channels without first restoring physical units. We preserve

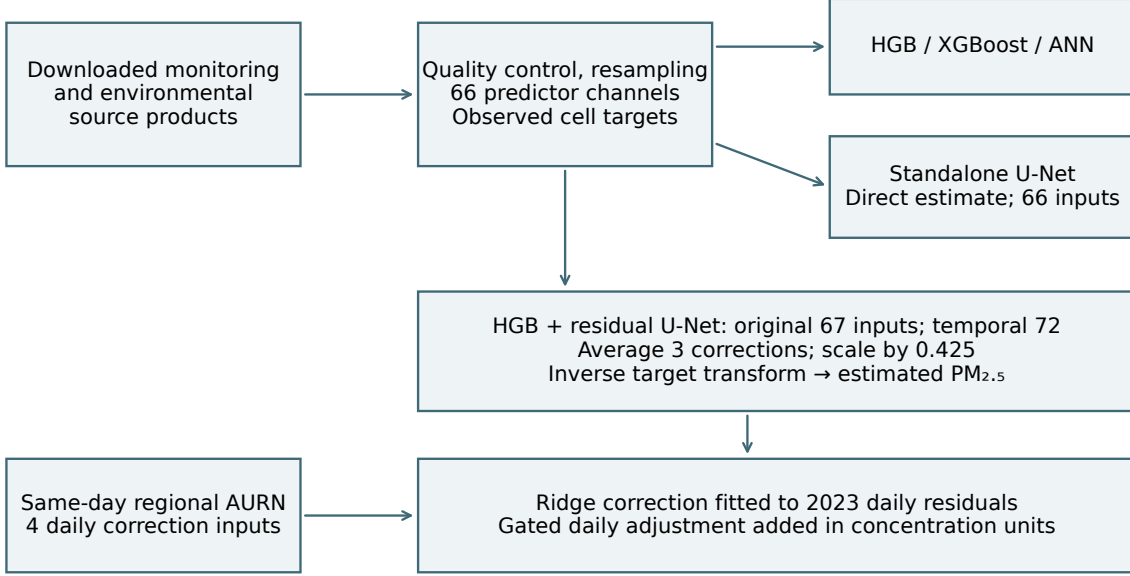


Figure 1: Implemented data and model paths. The diagram describes processing operations; it is not an experimental result. Unobserved cells do not acquire fabricated training labels. The hybrid additionally receives the HGB baseline map as its 67th input channel in the original configuration. The temporal extension inserts five history channels before the HGB map, giving 72 inputs. The AURN correction subsequently adjusts concentration-scale temporal outputs without changing these neural inputs. Its numerical audit uses station rows, not newly generated full-grid AURN maps.

those features to reproduce the recorded experiment, rather than silently changing its inputs. Its physical interpretation and the resulting model-status limitation are discussed in Section 7.5.

4 Methods

4.1 Model inputs and supervision

Let $X_t \in \mathbb{R}^{66 \times 48 \times 64}$ denote the retained predictor maps, m_{tg} indicate a labelled cell, and y_{tg} be its target. All original configurations use this retained feature contract. The temporal extension adds five history summaries to the U-Net only. The underlying models used for reported 2024 predictions fit on 2021–2023; the subsequent AURN correction fits separate 2023 out-of-year residuals. Tree and ANN models receive cell feature vectors; the U-Nets process daily grids. Final source weights are $s_{tg} = 1$ for LAQN-containing cells and $s_{tg} = 0.15$ for Breathe-only cells. This overrides the prepared Breathe-only weight of 0.35 for final training. The 4:1 mixed-target aggregation in Equation 1 is separate from this sample weighting.

4.2 Tabular boosting and ANN configurations

HGB and XGBoost each combine a model fitted on the raw target with one fitted in standardised log-target space. Define

$$T(y) = \frac{\log(1 + y) - \mu}{\sigma}, \quad T^{-1}(z) = \text{clip}(\exp(\mu + \sigma z) - 1, 0, 500), \quad (2)$$

where μ and σ are fitted using training labels. If $f_{\log}$ and f_{raw} are the two predictors, their concentration-scale blend is

$$\hat{y}^{\text{tree}} = \text{clip}\left[\lambda T^{-1}(f_{\log}(X)) + (1 - \lambda) \text{clip}(f_{\text{raw}}(X), 0, 500), 0, 500\right]. \quad (3)$$

The frozen log-model weights are 0.71 for unconstrained HGB, 0.68 for limited-monotonic HGB and 0.83 for XGBoost. The constrained HGB makes only the three IDW concentration-history features non-decreasing; it does not impose monotonic effects on every environmental feature. XGBoost uses 858 log-target trees and 115 raw-target trees, frozen from the preceding validation phase.

Both HGB variants use squared-error loss, 24 minimum samples per leaf and no internal early stopping. The log-target member uses 350 iterations, learning rate 0.055, 31 maximum leaf nodes and L2 regularisation 2.0; the raw-target member uses 450 iterations, learning rate 0.05, 63 maximum leaf nodes and L2 regularisation 3.0. The frozen XGBoost log/raw members use maximum depths 8/6, learning rates 0.025/0.03, minimum child weights 12/8 and L2 regularisation 6.0/3.0, respectively. The stored model bundles and training scripts retain the complete estimator configuration.

The ANN baseline is a scikit-learn multilayer perceptron with $66 \rightarrow 256 \rightarrow 128 \rightarrow 1$ units, ReLU hidden activations, Adam optimisation, regularisation parameter 10^{-4} , learning rate 5×10^{-4} and batch size 128. It uses training-only internal early stopping and five seeds (11, 22, 33, 44 and 55); predictions are combined across the five fitted members. This internal stopping procedure differs from the neural spatial models' year-based duration selection. The comparison therefore holds observation rows and fitting years constant, but does not equalise every training or tuning decision across model families.

4.3 Hybrid residual U-Net

The hybrid learns a spatial correction to unconstrained HGB in transformed target space. With $b_t = T(\hat{y}_t^{\text{HGB}})$, the U-Net receives $[X_t, b_t]$, giving 67 input channels, and predicts a residual map r_k for ensemble member k . The final map is

$$\hat{y}_t^{\text{hybrid}} = T^{-1} \left(b_t + \frac{\alpha}{3} \sum_{k=1}^3 r_k([X_t, b_t]) \right), \quad \alpha = 0.425. \quad (4)$$

The correction is added before the inverse transformation, not directly to $\text{PM}_{2.5}$ concentrations. The recorded final hybrid target moments are $\mu = 2.223419$ and $\sigma = 0.489477$.

The encoder widths are 32, 64 and 128, with a width-256 bottleneck. Residual convolution blocks combine 3×3 convolutions, Group Normalization, SiLU activations, spatial dropout and squeeze-and-excitation. Three decoder stages use bilinear upsampling and concatenated encoder skips, followed by a 1×1 output head. The base dropout is 0.12, scaled by depth in the code. The output head begins at zero so that the initial hybrid estimate coincides with its baseline. These choices specify the implemented architecture; individual performance gains from each block have not been isolated.

For residual training, the target is $T(y_{tg}) - b_{tg}$. A masked, weighted SmoothL1 loss is used with transition parameter one:

$$\mathcal{L} = \frac{\sum_{t,g} m_{tg} s_{tg} d(y_{tg}) \ell_1(r_{tg} - [T(y_{tg}) - b_{tg}])}{\max(1, \sum_{t,g} m_{tg} s_{tg} d(y_{tg}))}, \quad \ell_1(e) = \begin{cases} e^2/2, & |e| < 1, \\ |e| - 1/2, & \text{otherwise.} \end{cases} \quad (5)$$

Here $d(y)$ is a training-derived density weight obtained from a source-weighted, Gaussian-smoothed histogram and an inverse-square-root density transformation. Scaling and clipping enforce a source-weighted mean of one and bounds of 0.5 and 2.5. The sums in Equation 5 are implemented over each mini-batch. No residual-magnitude penalty or raw-scale auxiliary loss is active in this frozen configuration.

Training uses AdamW [19], initial learning rate 3×10^{-4} , weight decay 10^{-4} , gradient clipping at 5 and an exponential moving average with decay 0.98. Mini-batches contain eight days, with gradient accumulation over two batches. The final three seeds replay validation-selected

schedules: seed 42 for 16 epochs, seed 11 for 6, and seed 22 for 27. The corresponding residual predictions are averaged as in Equation 4. Training-period HGB maps are generated by an HGB fitted on the same 2021–2023 labels. This in-sample residual-stacking design must be distinguished from out-of-fold baseline training.

4.4 Standalone U-Net

The standalone model uses the same backbone pattern but receives only the 66 predictor channels. It directly estimates $T(y)$; it has no HGB input, baseline addition or residual target. Its single-network parameter count is 1,932,481. The direct-prediction loss uses Equation 5 with residual error replaced by $u(X) - T(y)$. Source and density weighting are retained, with training-only preprocessing statistics.

Duration selection fits 2021–2022 and evaluates 2023 with seed 42, a maximum of 60 epochs and patience 12. Epoch 25 and its learning-rate schedule are then frozen. Fresh seeds 42, 11 and 22 each fit 2021–2023 for that schedule, using batch size 8, accumulation over two batches, AdamW and EMA decay 0.98. The standalone ensemble averages the three transformed outputs before applying T^{-1} . Seed 42 is the pre-specified single-network result. Individual seeds 11 and 22 are retained as sensitivity information, not selected as the primary single model after viewing 2024.

4.5 Past-only temporal extension

The temporal extension gives the residual network a longer description of recent pollution without changing the U-Net backbone or the measured targets. It adds a three-day mean, seven-day mean, seven-day maximum, seven-day standard deviation and a recent-change feature [20]. Let h_t denote the prepared `pm25_lag1_idw_causal` map for target day t . This field already ends at day $t - 1$; it is a standardised, previously clipped predictor rather than an untransformed monitor reading. For each cell define

$$\bar{h}_t^{(k)} = \frac{1}{n_k(t)} \sum_{j=0}^{n_k(t)-1} h_{t-j}, \quad n_k(t) = \min(k, t + 1), \quad k \in \{3, 7\}, \quad (6)$$

where the prepared sequence starts at index zero. The maximum and population standard deviation use the same trailing seven-entry window. The fifth feature is h_t minus the mean of up to three preceding entries ($h_{t-1}, h_{t-2}, h_{t-3}$); it is zero at the first entry. Thus the stored “three-day trend” is a level difference, not a fitted linear slope. Shorter windows are used at startup, and history continues across year boundaries.

Each new channel is centred and scaled using only fitting dates and cells inside London. Original predictors and missing-history indicators are retained. The network input is $[X_t, H_t, b_t]$: 66 existing predictors, the five temporal summaries H_t , and the HGB map, for 72 channels. The original residual loss, backbone, seed schedules and multiplier $\alpha = 0.425$ are retained. No same-day measured $\text{PM}_{2.5}$ is added through this history construction; the retrospective availability limits of the environmental inputs still apply.

The temporal experiment refits a common HGB baseline for its two arms rather than reusing the original archived maps. Its raw/log blend is selected by 2023 validation RMSE and frozen for the final fit. The added summaries enter the U-Net, not HGB. Therefore the temporal model must be compared with its fresh matched control, not treated as a one-factor modification of the earlier published hybrid. Both final arms use seeds 42, 11 and 22 and the same 2021–2023 fitting window.

4.6 Regional correction of temporal-hybrid predictions

Regional observations can supply information about pollution episodes that earlier London measurements alone may not capture. The extension therefore adds a small day-level correction to saved temporal-hybrid predictions; no HGB or U-Net weights are retrained in this experiment [18]. For each 2023 date, the training response is the mean observed-minus-predicted LAQN residual across available stations. A standardised four-feature Ridge regression estimates this daily residual, giving equal weight to each day. Feature scaling is fitted within each training fold, then on all 2023 dates for the frozen final correction.

Let a_t contain the four AURN features, q_t be the maximum of its first three concentration-valued entries, and $f(a_t)$ be the fitted Ridge output in $\mu\text{g m}^{-3}$. The selected correction is

$$g(q_t) = \text{clip}\left(\frac{q_t - 8}{15 - 8}, 0, 1\right), \quad (7)$$

$$\hat{y}_{tg}^{\text{AURN}} = \text{clip}\left[\hat{y}_{tg}^{\text{temporal}} + 0.75 g(q_t) \text{clip}(f(a_t), -5, 15), 0, 500\right]. \quad (8)$$

The Ridge regularisation parameter is 1. The concentration-dependent gate reduces changes on low-regional-pollution days, and shrinkage limits the adjustment. Unlike the U-Net residual in Equation 4, this correction is added after decoding, directly in concentration units.

Every location receives the same adjustment on a given date. Apart from final clipping, this preserves within-day spatial differences and hotspot ordering. Thus this experiment tests adjustment of the regional pollution level, not new spatial detail learned by U-Net. Four features are sufficient for the selected correction; neither the eight-field preparation archive nor its availability flag is appended wholesale to the neural network.

5 Experimental protocol

5.1 Chronology and common evaluation rows

The original development protocol uses 2021–2022 for fitting and 2023 for external validation. The later three-year experiments refit configurations on 2021–2023 and score the available 2024 observations. The refit code does not use 2024 labels for parameter or scaler fitting, early stopping, epoch choice, blending or calibration. Nevertheless, earlier project runs had already revealed 2024 results. Absence of direct fitting on those labels is therefore not equivalent to an untouched evaluation or immunity from adaptive research decisions.

The requested prediction interval includes all 366 days of 2024, but station labels in the prepared package stop on 14 December. The primary common benchmark consequently contains 9,619 LAQN station-days on 349 dates, or 95.36% of that calendar year’s days. The final 17 days are unscored; no missing observations are generated to complete the year. Although the package also contains 131,754 Breathe station-days in this interval, the paper’s main results refer only to LAQN. Full-grid U-Net prediction arrays cover all 366 requested dates, including unscored dates.

5.2 Temporal comparison and model status

The temporal experiment first fits 2021–2022 and evaluates 11,102 LAQN station-days in 2023. Its pre-written promotion rule requires lower overall RMSE, higher R^2 , lower high-concentration MAE, non-increasing MAE below $15 \mu\text{g m}^{-3}$, and signed bias below $15 \mu\text{g m}^{-3}$ no greater than $\max(0.10, \text{control bias})$. It improves the error measures slightly but increases that bias from $+0.3892$ to $+0.4024 \mu\text{g m}^{-3}$, missing the guard by $0.0132 \mu\text{g m}^{-3}$. Despite this, the saved experiment subsequently evaluates both three-seed final fits on 2024. That comparison is an exploratory retrospective result, not a validation-approved promotion or an untouched confirmation. The original hybrid remains the official model.

5.3 AURN selection and verification protocol

The correction uses 11,102 saved 2023 LAQN station predictions from a temporal model fitted on 2021–2022. Twelve folds hold out one calendar month at a time. They are leave-one-month-out folds, not forward-only rolling-origin folds: later months can train a fold evaluated on an earlier month. Four Ridge penalties (0.1, 1, 10, 100), three lower gates (8, 10, 12), three upper gates (15, 20, 25) and four shrinkage factors (0.25, 0.5, 0.75, 1) yield 144 recipes per information source. Eligible recipes improve RMSE, R^2 and high-band MAE without worsening MAE below $15 \mu\text{g m}^{-3}$; the lowest-RMSE eligible recipe is selected. This rule does not contain the temporal model’s signed-bias guard.

The same-day source passes this rule; the lag-one alternative has no eligible recipe and is not advanced as an improved model. The implementation saves its selection before reading 2024 labels, refits the chosen correction on all 2023 daily residuals and applies it to the unchanged 2024 temporal predictions. This reuses a previously examined retrospective benchmark. It is neither an untouched confirmation nor a prospective availability test for ratified daily AURN data, and the selected cross-validation score is not an unbiased test score.

5.4 Metrics and descriptive uncertainty

For station observations y_i , predictions p_i and errors $e_i = p_i - y_i$, the primary metrics are

$$\text{MAE} = \frac{1}{N} \sum_i |e_i|, \quad \text{RMSE} = \sqrt{\frac{1}{N} \sum_i e_i^2}, \quad R^2 = 1 - \frac{\sum_i e_i^2}{\sum_i (y_i - \bar{y})^2}. \quad (9)$$

Mean error reports bias. These are pooled station-day metrics, so stations with more available days have greater representation. R^2 is not a percentage accuracy. Within-day spatial correlation is calculated after subtracting each day’s observed and predicted station means and pooling the centred values on days with at least three stations. It assesses relative within-day variation at monitors rather than agreement at unmonitored cells.

High-concentration diagnostics use $y_i \geq 25 \mu\text{g m}^{-3}$, a project-defined analysis threshold, not a claimed regulatory or health standard. The benchmark has 216 such rows. The saved comparison audit also provides 50,000 paired calendar-day bootstrap repetitions. Each resample preserves the paired models and all station rows within a sampled day; RMSE is recomputed on the sampled rows. Reported intervals are the 2.5th and 97.5th percentiles of model differences. They are descriptive, post-selection and unadjusted for multiple comparisons. Resampling whole days retains within-day clustering but does not model serial dependence across successive dates. Separately, the temporal report provides a paired seven-calendar-day block-bootstrap RMSE-difference interval, retaining short-range temporal grouping as well as paired rows. That interval and the original single-day intervals are distinct analyses and are labelled separately. The AURN comparison additionally recomputes a paired moving seven-calendar-day block interval using 5,000 resamples, keeping all station rows for each sampled day and seed 20260930. Negative corrected-minus-temporal RMSE differences favour the correction. None of these intervals accounts for prior benchmark inspection or establishes uncertainty in predictions at unmonitored cells.

5.5 Artifact verification

The original comparison tables and performance figures are generated from the saved station prediction table. The accompanying script checks the 9,619 unique station-day keys, date coverage and finite values, and independently recomputes MAE, RMSE, R^2 , bias, high-event errors and centred spatial correlation. Recomputed metrics agree with the saved experiment metrics within 10^{-6} . File hashes record the input artifacts. Bootstrap intervals are transcribed from the final saved audit, rather than recomputed during manuscript preparation. These checks

Table 2: Common LAQN benchmark: 9,619 station-days, 1 January–14 December 2024; final fitting on 2021–2023. MAE and RMSE are in $\mu\text{g m}^{-3}$. The later temporal comparison uses its own matched HGB refit. AURN adds same-day information and is separated from the unchanged-input comparisons. Bold identifies the best scores within each group, not statistical significance or model promotion.

Configuration	MAE ↓	RMSE ↓	R^2 ↑
<i>Original archived configurations</i>			
Hybrid HGB–U-Net (3 seeds)	2.3405	3.7765	0.5179
HGB (unconstrained)	2.3550	3.7920	0.5139
HGB (limited monotonic)	2.4344	3.8471	0.4997
XGBoost	2.3712	3.8739	0.4927
ANN (5 seeds)	2.5768	4.0871	0.4353
Standalone U-Net (3 seeds)	2.4891	4.1227	0.4254
Standalone U-Net (seed 42)	2.5963	4.2816	0.3803
<i>Later matched temporal comparison (experimental)</i>			
Matched non-temporal hybrid (3 seeds)	2.3443	3.7815	0.5166
Temporal hybrid (3 seeds; experimental)	2.3361	3.7679	0.5201
<i>Additional same-day regional information (experimental)</i>			
Temporal hybrid + AURN correction	2.1343	3.2309	0.6471

establish consistency of saved artifacts and arithmetic; they are not a new end-to-end retraining or independent validation of every provider measurement.

The earlier matched temporal comparison is sourced from its metrics CSV, report and audit [20]. The prior audit records recalculated metrics, identical evaluation rows, unchanged source inputs and hashes of all six final checkpoints. The supplied supplement now provides the temporal arm’s 2023 and 2024 station predictions, but not its neural checkpoints or the matched non-temporal control predictions. The old matched-pair interval remains a transcription, rather than a newly rerun temporal-versus-control bootstrap.

The AURN reproduction independently verifies all 11,102 selection rows and 9,619 benchmark rows against the original station archive, including unique station/date identities, grid positions and unchanged float32 observations [18]. It checks the feature and supplement hashes, reruns all recipe trials and the final Ridge fit, and independently recomputes the metrics and seven-day interval. Selection decisions agree exactly; the largest numeric difference across trial records is below 2×10^{-5} . Reproduced RMSE differs from the shared value by less than $7 \times 10^{-7} \mu\text{g m}^{-3}$, consistent with minor floating-point variation. This verifies the correction from supplied neural predictions, not independent regeneration of those predictions or a full end-to-end training audit.

6 Results

6.1 Common-benchmark performance

The AURN-assisted temporal hybrid has the lowest MAE and RMSE in Table 2, but receives additional same-day regional measurements. Its paired comparison is detailed in Section 6.3; the earlier temporal comparison appears in Section 6.2. Within the original configurations, the hybrid’s RMSE is $3.7765 \mu\text{g m}^{-3}$, compared with $3.7920 \mu\text{g m}^{-3}$ for unconstrained HGB, a relative reduction of approximately 0.41%. The small difference is visible against the broader configuration spread in Figure 2. The models are close enough that the distinction between numerical ranking and evidence of a reliable advantage matters.

The standalone ensemble reaches RMSE $4.1227 \mu\text{g m}^{-3}$, exceeding both HGB and the hybrid. Its pre-specified single member, seed 42, reaches $4.2816 \mu\text{g m}^{-3}$. The other two single networks obtain 4.1973 (seed 11) and $4.0692 \mu\text{g m}^{-3}$ (seed 22). Reporting the lower seed-22 result as a pre-

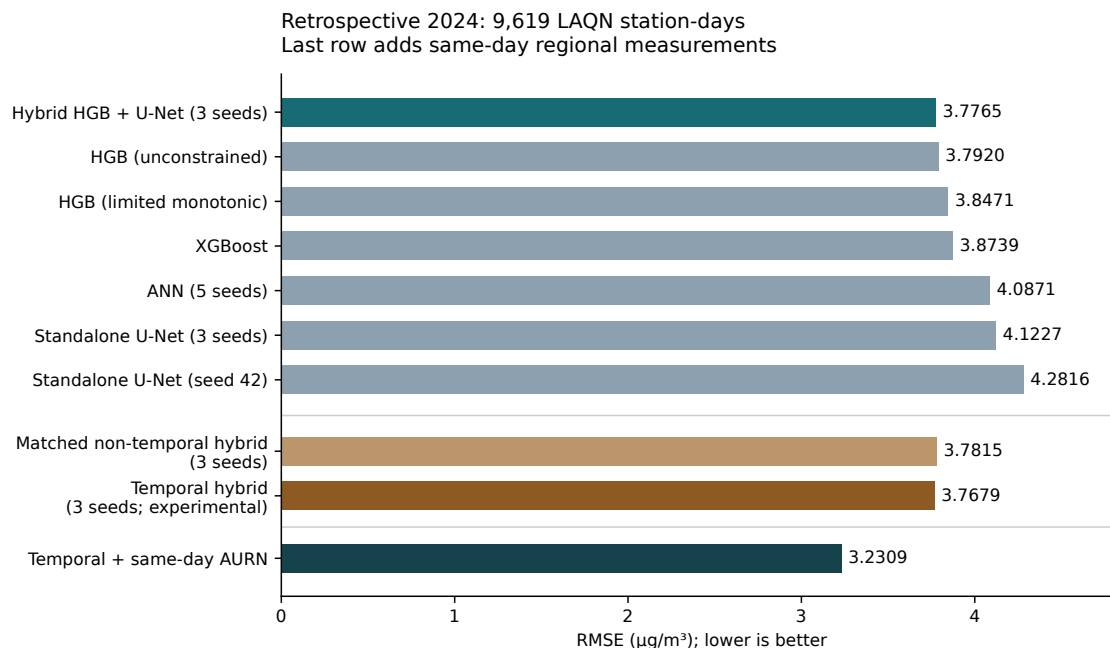


Figure 2: Original RMSE values recomputed from saved predictions, with the two later temporal-comparison values and the locally reproduced AURN result. All score 9,619 LAQN rows; the last row uses additional same-day measurements and is not a same-input architecture comparison. The horizontal axis starts at zero. Bars show point estimates, not uncertainty, computational cost or deployment status.

selected best standalone model would use evaluation outcomes to choose the model. These results characterise the tested training recipe and do not establish the best performance attainable by every standalone U-Net.

The tabular baselines also show metric-dependent ordering. XGBoost has lower MAE than limited-monotonic HGB but higher RMSE. The ANN has higher RMSE than the tree configurations, while its RMSE is slightly below the standalone U-Net ensemble. Thus, a single statement that all neural models outperform all tree models would contradict the observed results.

6.2 Temporal result and its matched comparison

The temporal hybrid obtains MAE $2.3361 \mu\text{g m}^{-3}$, RMSE $3.7679 \mu\text{g m}^{-3}$ and $R^2 = 0.5201$. Its freshly fitted non-temporal control obtains $2.3443 \mu\text{g m}^{-3}$, $3.7815 \mu\text{g m}^{-3}$ and 0.5166 , respectively. The matched RMSE reduction is $0.0136 \mu\text{g m}^{-3}$ (approximately 0.36%), with MAE lower by $0.0082 \mu\text{g m}^{-3}$ and R^2 higher by 0.0035 . The saved paired seven-day-block interval for temporal-minus-control RMSE is $[-0.0203, -0.0075] \mu\text{g m}^{-3}$. This describes a small retrospective gain in this experiment, not a prospective superiority claim. It is not an interval for comparison with the original $3.7765 \mu\text{g m}^{-3}$ hybrid or with HGB alone.

The high-concentration MAE falls only from 14.1090 to $14.0490 \mu\text{g m}^{-3}$ (Table 3). Thus the lower overall RMSE does not resolve the dominant episode-error problem. The temporal configuration is the stronger of that experiment’s two configurations, but remains experimental because it did not pass the earlier 2023 promotion guard. This qualification is part of interpreting its favourable 2024 scores, not a reason to discard those scores.

6.3 Reproduced AURN-assisted result

The same-day regional correction reduces temporal-hybrid RMSE from 3.7679 to $3.2309 \mu\text{g m}^{-3}$ (14.25%) and MAE from 2.3361 to $2.1343 \mu\text{g m}^{-3}$ (8.64%). R^2 increases from 0.5201 to 0.6471 . The reproduced seven-day-block interval for corrected-minus-temporal RMSE is $[-0.8294,$

Table 3: Temporal comparison on the same 2024 observations. All values are in $\mu\text{g m}^{-3}$; concentration bands are defined by observed values. Both models remain strongly biased low on high-concentration observations.

Diagnostic	Matched control	Temporal hybrid
Overall signed bias	-0.7759	-0.7685
MAE, observed < 15	1.7239	1.7174
Bias, observed < 15	-0.0601	-0.0549
MAE, observed $15 \leq y < 25$	5.9552	5.9437
MAE, observed ≥ 25	14.1090	14.0490
Bias, observed ≥ 25	-14.1090	-14.0490

Table 4: Locally reproduced regional-correction diagnostics on unchanged 2024 observations. Errors and bias are in $\mu\text{g m}^{-3}$. The bands contain 8,625, 778 and 216 observations, respectively.

Diagnostic	Temporal hybrid	With AURN correction
Overall signed bias	-0.7685	-0.4832
MAE, observed < 15	1.7174	1.6967
MAE, observed $15 \leq y < 25$	5.9437	4.7401
MAE, observed ≥ 25	14.0490	10.2219
Bias, observed ≥ 25	-14.0490	-10.1787

$-0.2373] \mu\text{g m}^{-3}$. This is a larger retrospective gain than the temporal-history extension, but it also changes the available information. It is not evidence that a different U-Net architecture alone improves performance by 14.25%.

High-concentration MAE falls from 14.0490 to 10.2219 $\mu\text{g m}^{-3}$ (27.24%), while MAE below 15 $\mu\text{g m}^{-3}$ decreases from 1.7174 to 1.6967 $\mu\text{g m}^{-3}$. These results show that the pooled gain does not come at the expense of increased mean absolute error on ordinary observations. However, 213 of 216 high readings remain underpredicted. On 11 March 2024, the mean across 33 evaluated stations is 34.76 $\mu\text{g m}^{-3}$; the correction raises the mean prediction from 11.17 to 15.83 $\mu\text{g m}^{-3}$. That day’s MAE improves from 23.60 to 18.94 $\mu\text{g m}^{-3}$, leaving a large unresolved episode error (Figure 3).

6.4 Magnitude and uncertainty of the original hybrid improvement

Table 5 reports the saved paired-day bootstrap comparisons for the original five-configuration refit. For unconstrained HGB, the RMSE difference is 0.01548 $\mu\text{g m}^{-3}$. Its descriptive interval runs from -0.00003 to 0.02830 $\mu\text{g m}^{-3}$. The intervals for limited-monotonic HGB and XGBoost also include zero. These intervals do not establish a clear RMSE advantage of the hybrid over those alternatives, but neither do they prove equivalence. The ANN interval excludes zero in this descriptive analysis; it remains subject to the same post-selection and dependence limitations.

The later standalone audit reports a standalone-ensemble-minus-HGB RMSE difference of

Table 5: Saved descriptive paired-day bootstrap results: candidate RMSE minus hybrid RMSE, in $\mu\text{g m}^{-3}$. Positive differences favour the hybrid. The near-zero HGB endpoint is shown to five decimals to avoid hiding its negative sign.

Candidate	Difference	Descriptive 95% interval
HGB (unconstrained)	0.01548	$[-0.00003, 0.02830]$
HGB (limited monotonic)	0.07056	$[-0.00158, 0.14843]$
XGBoost	0.09739	$[-0.00419, 0.19902]$
ANN (5 seeds)	0.31053	$[0.06714, 0.54546]$

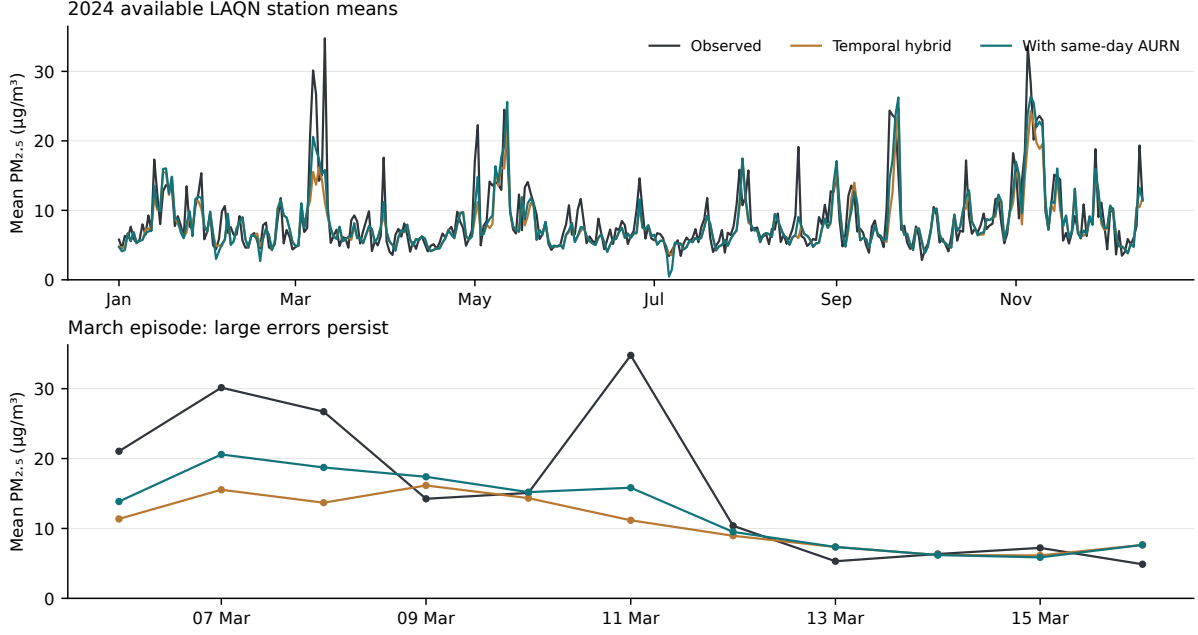


Figure 3: Daily means over the same available LAQN station rows: observed values, temporal predictions and AURN-corrected predictions. Top: the 349 labelled benchmark dates. Bottom: the March episode. Station availability varies by date, so these are monitoring-network means rather than London area-weighted means. No labels are invented for the unscored final 17 days.

Table 6: Original-configuration diagnostic results on the same LAQN rows. Bias and high-concentration MAE are in $\mu\text{g m}^{-3}$. High-concentration MAE uses the 216 observations with measured PM_{2.5} at least $25 \mu\text{g m}^{-3}$; spatial r is pooled within-day-centred correlation.

Configuration	Bias	Spatial $r \uparrow$	High MAE $\downarrow$
Hybrid HGB-U-Net (3 seeds)	-0.7705	0.6537	14.0668
HGB (unconstrained)	-0.8127	0.6337	14.2033
HGB (limited monotonic)	-0.8028	0.6579	14.5715
XGBoost	-0.8956	0.6470	15.2875
ANN (5 seeds)	-0.7890	0.6281	14.7872
Standalone U-Net (3 seeds)	-1.3367	0.6441	16.2035
Standalone U-Net (seed 42)	-1.3095	0.6228	16.8923

$0.3306 \mu\text{g m}^{-3}$ with descriptive interval $[0.0926, 0.5712] \mu\text{g m}^{-3}$, and a standalone-minus-hybrid difference of $0.3461 \mu\text{g m}^{-3}$ with interval $[0.1029, 0.5908] \mu\text{g m}^{-3}$. These use the same day-resampling approach and do not convert the retrospective comparison into a confirmatory hypothesis test.

6.5 Spatial variation and high-concentration errors

All reported configurations have negative mean error (Table 6). The original hybrid’s within-day spatial correlation is 0.6537, above unconstrained HGB’s 0.6337, but below limited-monotonic HGB’s 0.6579. This shows that the model with the lowest pooled RMSE does not lead every spatial diagnostic. The correlations remain measurements at available stations, with no implication that each intervening grid cell is validated.

High-concentration error remains substantial for the hybrid and HGB, at 14.0668 and $14.2033 \mu\text{g m}^{-3}$ MAE, respectively. These 216 rows represent only 2.25% of the benchmark but account for approximately 39.0% of HGB’s total squared error. The parity plots in Figure 4 show the compression of predictions at higher observed concentrations. The standalone

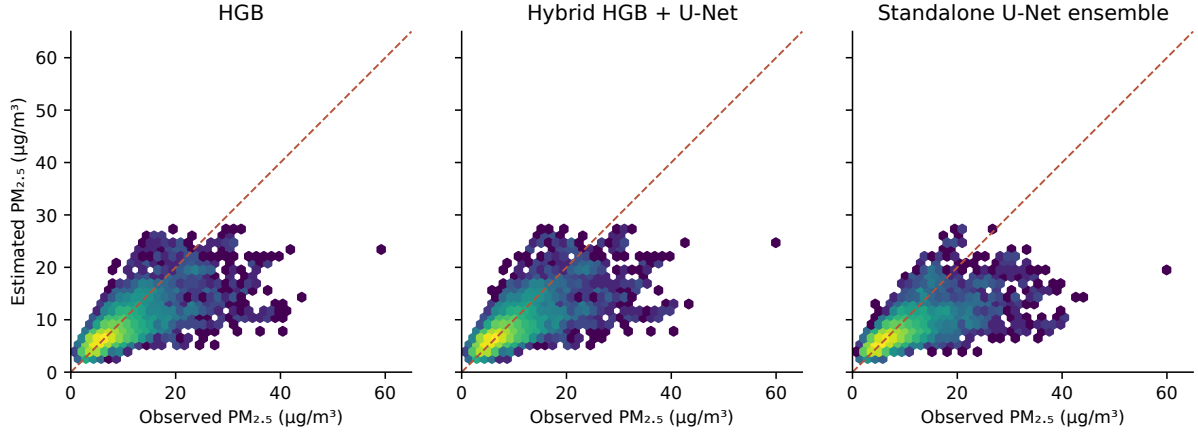


Figure 4: Saved station predictions against measured $\text{PM}_{2.5}$ for three original configurations, not the temporal extension, on all 9,619 common rows. Hexagonal bins use logarithmic counts to reveal concentration density; colour intensities are independently normalised per panel and are not compared across panels. Axes are common and the dashed line denotes perfect agreement. These are station comparisons, not full-grid ground-truth maps.

ensemble has high-concentration MAE $16.2035 \mu\text{g m}^{-3}$. Its threshold recall is $1/216$ (0.46%), whereas HGB and the hybrid each flag $12/216$ (5.56%) of these observed high rows at the same threshold. These are retrospective threshold diagnostics, not a deployed event-warning evaluation.

7 Discussion and limitations

7.1 What the model comparison establishes

The results support retaining HGB as a strong reference for this problem and describing the hybrid as a modest numerical refinement. The U-Net correction can use neighbouring predictor patterns, but the baseline already incorporates coordinates, environmental information and prior-day IDW fields. Limited additional information after baseline fitting is a plausible explanation for the small gain; it is an interpretation, not a demonstrated causal mechanism. The additional temporal summaries give a further small matched gain, while leaving high-concentration errors largely unresolved. No controlled ablation isolates the contribution of every original input group, architectural block, ensemble size or training-loss choice in the final three-year experiment.

The AURN comparison supplies a more substantial improvement by adding regional measurements on the target day. With the temporal predictions fixed, the paired comparison isolates the effect of the combined regional-feature/Ridge correction stage, not the contribution of each regional feature separately. Its daily scalar adjustment cannot establish improved within-day hotspot ordering. A stronger information set and a different learning architecture are distinct explanations and should not be conflated.

The standalone comparison is informative but not a pure one-factor ablation of the HGB input. Removing HGB also changes the learning target from a residual to a direct prediction, and the frozen training durations differ. Similarly, the three-seed spatial models and five-seed ANN do not have matched ensemble budgets. The study does not measure inference latency, energy use or a controlled cost–accuracy trade-off. A preference for the simpler HGB pipeline is therefore a design judgement supported by close error scores, not a measured runtime superiority claim.

7.2 Temporal evaluation and information availability

Prior inspection of 2024 is the main barrier to a prospective performance claim. Preserving frozen refit scripts and hashes prevents silent changes to the reported configurations but cannot erase knowledge gained from previous evaluation runs. A future untouched period or nested rolling-origin protocol is required to assess robustness to research decisions and temporal change. The present package also leaves the last 17 days of 2024 unscored.

Operational availability is a second, separate constraint. Same-day ERA5 reanalysis and same-quarter Sentinel-2 composites are appropriate for retrospective reconstruction, but may be unavailable at a real forecast issue time. Census and emissions inventories are static reference products rather than contemporaneous daily observations. The AURN correction also requires same-day ratified daily measurements, with at least 18 valid hours. The experiment does not show that these inputs are available by a forecast issue time. A deployment-oriented evaluation would need covariates available by a specified issue time, archived forecast weather and explicitly lagged satellite composites. No such evaluation is reported here.

7.3 Spatial supervision and source dependence

The benchmark evaluates temporal behaviour at available regulatory stations; it is not a geographically blocked holdout. Earlier observations from the same sites can influence later IDW predictors. Some station-day observations also share grid cells, so row count is not a count of independent spatial locations. The 1,719-cell prediction domain and visually smooth maps do not provide full-grid truth. Regional transfer, accuracy at previously unmonitored locations and improvement at specific neighbourhoods remain unestablished.

Breathe observations provide much of the training coverage, but source weighting does not establish that their residual errors are negligible. Their documented reference-network correction also means that sensor provenance should not be treated as wholly independent of LAQN. This study does not independently audit the timing or reference data used by the upstream sensor calibration system, and therefore does not make an end-to-end independence claim. The 4:1 mixed-cell target and 0.15 Breathe-only training weight are specific modelling decisions, not universally validated reliability ratios.

7.4 Stacking, preprocessing and statistical limits

In-sample HGB predictions used during residual training can differ in error distribution from predictions on later data. An out-of-fold or time-blocked baseline would better align residual training with generalisation error. The current design contains no direct 2024-label fitting in the refit, but this does not remove the stacking limitation. Likewise, normalising the existing float16 cube preserves its earlier clipping and quantisation.

The original day-level bootstrap omits across-day serial dependence. The temporal comparison’s seven-day blocks address short-range dependence, but neither analysis corrects for multiple comparisons or benchmark inspection. Intervals near zero should be interpreted cautiously, and they are not spatial uncertainty bands for the output maps. High-event underprediction is observed, but attribution to log targets, robust loss, density weighting, distribution shifts or smooth covariates would require additional controlled experiments. These possibilities are hypotheses for follow-up work.

7.5 Regional-feature implementation and verification limits

The wind-direction feature requires correction before a physically interpreted deployment. The supplied code applies $\text{atan2}(-u, -v)$ to standardised rather than physical ERA5 components. In a diagnostic copy, undoing the recorded scales changes inferred direction by a median 10.52

degrees, with changes above 30 degrees on 202 of 1,461 dates. The direction-weighted concentration changes by $0.396 \mu\text{g m}^{-3}$ on average and up to $16.343 \mu\text{g m}^{-3}$. These are feature differences, not measured changes in model error. Inverse scaling also cannot undo the cube’s clipping and float16 quantisation. A separately labelled experiment using original physical winds is needed; the numerical results here deliberately reproduce the supplied features unchanged.

This defect does not show that the observed improvement is spurious: the regional mean and maximum remain useful candidate information. It does mean that the directional feature cannot be described as physically validated. Likewise, reproducing correction arithmetic from shared station predictions does not regenerate the upstream temporal U-Net or demonstrate full-grid accuracy. These verification limits, the upstream temporal bias-guard failure and retrospective information availability preclude automatic promotion of the AURN-assisted model to the official dashboard.

8 Conclusion

A completed multi-source Greater London pipeline enables a transparent comparison of daily $\text{PM}_{2.5}$ estimation configurations on a shared regulatory benchmark. Among the original models, the hybrid reaches RMSE $3.7765 \mu\text{g m}^{-3}$, compared with HGB’s $3.7920 \mu\text{g m}^{-3}$ and the tested standalone U-Net ensemble’s $4.1227 \mu\text{g m}^{-3}$. Past-only temporal summaries provide a small further gain. The strongest reproduced numerical result adds a same-day regional AURN correction to the temporal hybrid: MAE $2.1343 \mu\text{g m}^{-3}$, RMSE $3.2309 \mu\text{g m}^{-3}$ and $R^2 = 0.6471$ on 9,619 unchanged station-days. Relative to the temporal hybrid, RMSE falls by 14.25%, with descriptive seven-day-block interval $[-0.8294, -0.2373] \mu\text{g m}^{-3}$ for corrected minus control. High-concentration MAE falls by 27.24% to $10.2219 \mu\text{g m}^{-3}$, but 213 of 216 high readings remain underpredicted. Same-day regional information improves reconstruction without demonstrating that U-Net has learned finer hotspots.

The findings justify a bounded conclusion about this implementation and benchmark, not a claim of general U-Net superiority, state-of-the-art performance, operational forecasting or validated accuracy throughout every grid cell. The original hybrid remains the official configuration; the wind feature needs correction and a separately audited comparison before promotion. The next substantive evaluation should combine an untouched time period, spatially blocked assessment, issue-time-available covariates and out-of-fold residual baselines. Those are proposed experiments and are not reported as completed work.

Data and code availability

This manuscript is based on the local `pm25_london_bundle` implementation and its saved experiment artifacts. The accompanying manuscript package includes LaTeX source, bibliography, generated tables, figures, the asset generation script and an evidence manifest. Raw data, fitted model binaries and the full processed cube are not bundled with the manuscript. Provider references identify the original sources; Breathe API access requires an appropriate key. Project code and curated result reports are maintained in [the Aitken repository](#); the temporal experiment and aggregate audit are identified in Aitken project contributors [20]. The AURN import, reproduced selection/correction and verification boundaries are documented in Aitken project contributors [18]; supplied prediction archives remain local rather than being represented as public neural checkpoints. The maintained manuscript source is in `research_paper` within that repository. No archived data DOI or unrestricted raw-data redistribution licence is claimed. Provider terms and data access remain separate from code availability.

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A Artifact provenance and reproduction

The paper is based on completed saved runs, rather than newly trained models. Within `code/pm25_london_bundle/artifacts`, the principal directories are:

- `london_3year_tree_refit_20260905_141738`: HGB and XGBoost refits.
- `london_3year_unet_refit_20260905_142452`: hybrid refit.
- `london_exact_sklearn_ann_train2021_2023_test2024`: ANN refit.
- `london_3year_final_comparison_20260905_145050`: original common comparison, metrics and final bootstrap audit.
- `london_standalone_unet_20260906`: standalone experiment and extended common prediction table.

The paper’s `build_assets.py` recomputes its tables from the last directory’s `test_2024_laqn_all_models.csv.gz`. The original five-model comparison supplies the bootstrap table. Input SHA-256 values and verification scope are recorded in `evidence/provenance.json`; unrounded recomputed values appear in `evidence/recomputed_metrics.csv`. Run `py -3.10 research_paper/build_assets.py` from the Aitken repository root to regenerate the original comparison assets. An explicit `-bundle` path can point to the read-only original London bundle. The script also generates the temporal table from `results/temporal_model_20260912/metrics.csv`, checks the recorded `audit.json`, and records those files and `REPORT.md` in `evidence/temporal_provenance.json`. The `-temporal-only` option refreshes only the temporal tables and provenance without requiring the large original artifacts. Neither mode retrains models. The seven-day temporal interval is transcribed from the versioned report; it is not substituted for the original day-level intervals.

The AURN reproduction reads the two supplied `B_temporal_predictions.csv.gz` tables under `artifacts/temporal_tail_correction_20260912/`, and the separate `data/external_features/aurn/` inputs. The audit script `scripts/reproduce_aurn_supplement.py` preserves the supplied records and writes separate local results under `artifacts/aurn_reproduction_20260930/`. Curated checksums, observation identity checks and reproduced scores are versioned under `comparison/aurn_import_20260930/`. The manuscript’s `evidence/aurn_provenance.json` records the inputs used to regenerate its regional-correction tables and daily-mean figure. No HGB or U-Net model was retrained for this reproduction; only the small Ridge selection/refit was repeated. The AURN interval, unlike the earlier original/temporal intervals, was independently recomputed from predictions.

The handoff records a Python 3.10.5 environment with PyTorch 2.10.0+cu126, scikit-learn 1.7.2, XGBoost 3.1.2 and NumPy 2.2.6. These are recorded project environment versions, not a claim that the manuscript independently rebuilt each model in that environment. Full refit commands and preprocessing provenance remain in the model directories and project handoff.

B Ordered retained feature schema

The following names are copied from the actual prepared schema after removing the five fire/smoke channels. There are 66 retained channels; the hybrid concatenates its separately generated HGB map as channel 67.

The temporal extension inserts the following ordered channels before HGB:

67. `history_mean_3d`
68. `history_mean_7d`
69. `history_max_7d`
70. `history_std_7d`
71. `history_trend_3d`

Its separately generated HGB map is channel 72. The table below lists the unchanged 66 predictors shared with the original configurations.

Table 7: Retained predictor channels in model order.

Index	Stored feature name
1	gla_mask
2	gla_fraction
3	x_coordinate
4	y_coordinate
5	population_log_density
6	population_available
7	os_road_coverage
8	os_major_road_coverage
9	os Greenspace_fraction
10	os_tile_available
11	laei_log_pm25_all_2022
12	laei_log_pm25_road_2022
13	laei_available
14	elevation_mean_m
15	elevation_relief_m
16	elevation_available
17	era5_t2m_mean_c
18	era5_t2m_min_c
19	era5_t2m_max_c
20	era5_d2m_mean_c
21	era5_surface_pressure_hpa
22	era5_u10
23	era5_v10
24	era5_cloud_cover
25	era5_boundary_layer_height
26	era5_precip_mm
27	era5_solar_mj_m2
28	era5_wind_speed
29	era5_relative_humidity
30	era5_log_ventilation
31	era5_precip_3day_sum
32	era5_wind_speed_3day_mean
33	era5_boundary_layer_3day_mean
34	era5_relative_humidity_3day_mean
35	era5_rolling_window_days
36	s5p_no2
37	s5p_no2_available
38	s5p_no2_valid_fraction
39	s5p_no2_age_days
40	s5p_co
41	s5p_co_available
42	s5p_co_valid_fraction
43	s5p_co_age_days
44	s5p_aerosol_index
45	s5p_aerosol_index_available
46	s5p_aerosol_index_valid_fraction
47	s5p_aerosol_index_age_days
48	s5p_cloud_fraction
49	s5p_cloud_fraction_available
50	s5p_cloud_fraction_valid_fraction
51	s5p_cloud_fraction_age_days
52	s2_ndvi
53	s2_ndbi
54	s2_available
55	calendar_doy_sin
56	calendar_doy_cos
57	calendar_dow_sin
58	calendar_dow_cos
59	calendar_time_years
60	pm25_lag1_idw_causal
61	pm25_lag2_idw_causal
62	pm25_lag3_mean_idw_causal
63	pm25_lag1_nearest_distance_km
64	pm25_lag1_direct_observed
65	pm25_lag1_available
66	pm25_lag1_log_observation_count